Shu Qiuhao

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Education Experience

Northwestern Polytechnical University (985 Project) **Master of Science in Computer Science and Technology** 2023.09 - 2026.06

- **Average score**:90.05 (top 10%) First-class Academic Scholarship, Tencent Second-class Scholarship (2025)
- The student published three papers on multi-agent collaborative perception related code, and published related patents.
- subentry： LLM assistant,automatic push of public account,GAN trainer,image generation LoRA trainer and publish relevant software copyright.

Northwestern Polytechnical University (985 Project) **Computer Science and Technology, Bachelor's Degree** 2019.09 - 2023.06

- **GPA**: 3.77/4.0 (**top 15%)**to qualify for graduate school recommendation, CET-6 (602)
- **prize：**
  - University-level Second-Class Scholarship, Academic Excellence Award (2020), Outstanding Graduate (2023)
  - Self-reliant and Advanced Individual, Outstanding College Student, "Internet Plus" Innovation and Entrepreneurship Competition Excellence Award (2022)

Specialized skill

- **Basic knowledge:**Solid computer basic knowledge, Java, C++ and other languages, familiar with common data structures and algorithms.
- **Java:**Proficient in Java fundamentals including the Java Collection Framework, generics, and reflection. Able to implement common design patterns such as the Singleton pattern and Factory pattern in projects. Familiar with multithreading and concurrent programming, with understanding of thread pools, synchronized lock optimization, and the principles of volatile keyword.
- **JVM:**Understand the memory structure of JVM, garbage collection mechanism, GC algorithm, class loading mechanism, etc.
- **MySQL:**Proficient in MySQL, understanding indexes, locking mechanisms, logs, and transactions, and able to optimize SQL statements in projects.
- **Redis:**Master Redis usage and understand its underlying data structures, persistence mechanisms, and distributed cluster deployment. Be able to resolve cache penetration and cache breakdown issues under high concurrency in projects.
- **Framework:**Proficient in developing applications using mainstream frameworks such as Spring, SpringBoot, and MyBatis. Mastering Spring's AOP, IoC, Bean lifecycle, and SpringBoot auto-injection principles.
- **AI ability:**Master the basic principles of large models and use PyTorch for training and tuning. Have experience in model training, fine-tuning and application development, and can build large model applications in projects with tools such as Ollama.

Internship experience

Intelligent Chat Assistant Based on RAG 2025.03 - 2025.06

Project Description: This integrated chat platform combines instant messaging with AI-powered assistants, featuring friend management, real-time conversations, and document upload capabilities to build personalized knowledge spaces. The platform incorporates RAG-based knowledge-enhanced Q&A and external tool integration, delivering efficient knowledge management and intelligent assistant functions while supporting daily social interactions.

Technology stack: SpringBoot, MySQL, Redis, RabbitMQ, MyBatis, Netty, Elasticsearch, MinIO, Spring-Security

Core responsibilities:

- The system implements high-performance instant messaging services through a custom WebSocket protocol based on Netty. Redisson is employed to cache session lists and route users, optimizing loading performance while **maintaining user state in cluster environments**. Additionally, an **incremental offline message retrieval** mechanism ensures message order and integrity upon user reconnection.
- RedisBitmap is used to record the uploaded successful shards, and MinIO storage service is combined to achieve **document shard upload and breakpoint resumption**. Meanwhile,**RabbitMQ asynchronous notification** is used to decouple the file upload, parsing, and vectorization processes to ensure fast response of the interface.
- We built an RAG retrieval pipeline based on Elasticsearch, which achieved multi-channel recall by **integrating KNN vector retrieval and BM25 keyword matching**. The Cross-Encoder rearrangement model was introduced to optimize semantic alignment accuracy, and the vertical domain Q&A accuracy was improved by more than 25% while balancing system response performance.
- The hierarchical indexing scheme based on document tree structures **employs a "chapter-paragraph-sentence" hierarchy** with sentences as the smallest vector units, effectively resolving semantic chain fragmentation caused by conventional rigid slicing. Simultaneously,**the integration of parent-child indexing strategies** enables "fine-grained retrieval and broad-grained recall", addressing information fragmentation issues in long-document Q&A systems.
- The **LLM-based adaptive retrieval module** is designed to optimize initial intent recognition through reference resolution and semantic expansion. For scenarios where the ranking score falls below the threshold, the system **triggers query rewriting and rollback mechanisms** to enhance retrieval recall quality. Additionally, it integrates role and organizational labels to enable fine-grained permission control, ensuring data isolation and security.
- The system integrates APIs from multiple platforms to support various large models, utilizing WebSocket and WebFlux to enable **real-time responses from large models** and reduce initial response time. It maintains multi-round conversation contexts through Redis and introduces a summary memory strategy for ultra-long conversations. By dynamically compressing early-stage contexts, the system ensures conversational coherence while significantly optimizing inference costs.
- Based on the LangChain idea, the **autonomous hierarchical agent architecture** is designed, including basic Agent, task decomposition Agent and execution Agent, which improves the ability and scalability of the intelligent assistant. At the same time, external tools such as search and date are integrated for the model to call.

Project Experience

intelligent project management system 2025.05 - 2025.08

Project Description: Designed for laboratories and research institutions, this platform provides a streamlined management system covering the entire project lifecycle, enabling project creation, task allocation, and review

Batch processing, permission control, and message notification address issues in traditional scientific research management, including decentralized workflows, low approval efficiency, and cross-system collaboration difficulties.**Technology stack:**Spring Boot, Spring Cloud, Redis, MySQL, Nacos, OpenFeign, Sentinel

Core responsibilities:

- **Customized TransmittableThreadLocal (TTL)-based context management component** addresses identity loss scenarios in microservice chains caused by asynchronous thread pools and Feign cross-service calls. The request interceptor enables context-transparent traversal, reducing redundant database queries by over 60% through end-to-end information sharing. An integrated session auto-renewal mechanism ensures security.
- **To address the data consistency issue in cross-service approval processes, we introduced the Seata AT model to implement distributed transaction management**, which resolves the atomicity of multi-service write operations and ensures strong consistency of approval status under high concurrency and abnormal scenarios.
- **Cache Aside mode** is adopted, combined with cache invalidation strategy and hot data optimization, to solve the consistency problem between cache and database, effectively reduce the database pressure, and improve the system throughput in high concurrency scenarios.
- The approval process is restructured based on RocketMQ to adopt an asynchronous event-driven architecture, decoupling project services from workflow services. By implementing **message idempotence control, retry mechanisms, and dead-letter queue design**, the system ensures reliable message processing and integrates with WeCom for real-time notifications.
- OpenFeign + Sentinel is used to implement **service circuit breaking, degradation and flow limiting policies**, and Gateway + Sentinel is used to build a unified traffic governance system to effectively deal with sudden traffic shocks and ensure the overall availability of the system to reach 99.9%+.
- Customize Spring Cloud Gateway global filters to **enable unified authentication, black/white list control, and API latency tracking**. By implementing a token bucket rate-limiting algorithm with Redis and Lua scripts, the system achieves an average response time of 70ms, throughput of 1700 QPS, and 100% rate-limiting accuracy in a 100,000-traffic load test scenario.

Procoder Coding Community **individual event** 2025.10 - 2026.01

Project Overview:A knowledge-sharing and communication platform for programmers, supporting content creation, interactive comments, message notifications, QR code login, and user behavior analysis.Website Link

Technology stack: SpringBoot, MySQL, Redis, MyBatis, MyBatis-Plus, RabbitMQ

Core responsibilities:

- The design is based on a **distributed authentication system using JWT and Redis**, introducing a dual Token mechanism (Access Token and Refresh Token) to achieve Invisible refresh and session renewal reduce user re-login rate;**ThreadLocal manages user context** to reduce repeated queries and improve interface performance.
- **Using SSE to build a real-time feedback mechanism for WeChat QR code login**, combined with the state mapping model to achieve smooth synchronization of the scanning, following, and login processes; while reducing server I/O overhead, it ensures real-time synchronization of user states.
- **A parallel export scheme is designed** for large-scale PV/UV data export scenarios. Based on FastExcel+, custom thread pool and CountDownLatch, the task is split and processed concurrently, which significantly reduces the export time and improves the performance by about 20 times.
- **Build a multi-level caching system**(Caffeine + Redis) to optimize hot data access paths, significantly reducing database access pressure in a 2-core, 2GB environment. The system achieves stable throughput of approximately 1000 QPS.**A user activity ranking model is designed based on Redis ZSet**, with idempotent control and event mechanisms ensuring data consistency to enable low-latency updates for high-frequency behaviors.
- **High-frequency user behaviors such as comments, likes, and favorites are processed as events**, and asynchronous decoupling is implemented through RabbitMQ to reduce the latency of the main link. A reliable message consumption mechanism is designed to ensure the eventual consistency of message processing.
- **We established a unified logging and trace system**, implementing request-level traceability and issue localization through TraceID and AOP. By integrating Prometheus and Grafana, we achieved **system operational observability**. JMeter tests demonstrated that the query could process up to 700 HTTP requests per second within 10 minutes on 50 threads.

Scientific Research Task

Multi-agent collaborative perception **The National Natural Science Fund** 2024.03 - 2025.07

Paper Overview: This study investigates collaborative perception enhancement for multi-agent systems (e.g., vehicles and road infrastructure) through simulation and real-world datasets, addressing challenges like positioning noise, latency, and bandwidth constraints. By deploying the model in autonomous vehicle fleets, the system leverages information exchanged with neighboring vehicles to expand detection coverage, achieving fundamental collaborative perception capabilities.

Output results:

- J. Chen,**Q. Shu**, Y. Lu, Y. Zhang, and Y. Wang, "QCTF: A Quantized Communication and Transferable Fusion Framework for Multi-Agent Collaborative Perception," in IEEE Transactions on Intelligent Transportation Systems, doi: 10.1109/TITS.2025.3574725. (**First author, SCI Zone 1)**
- **Q. Shu**, J. Chen, Y. Lu, Y. Zhang, and Y. Wang, "CoRange: Collaborative Range-aware Adaptive Fusion for Multi-Agent Perception," in IEEE Transactions on Intelligent Vehicles, doi: 10.1109/TIV.2024.3478756. (**First author, SCI Zone 1)**
- J. Chen,**Q. Shu**, Y. Lu, Y. Zhang, and Y. Wang, "CEST: Enhancing Multi-Agent Perception via Communication-Efficient Spatial-Temporal Fusion," in IEEE Transactions on Intelligent Transportation Systems, doi: 10.1109/TITS.2026.3658043. (**First author, SCI Zone 1)**

Personal Summary

- Understand the basic principles and related application technologies of large models and generative AI, including Transformer, RAG, and MCP. Be able to quickly master new technologies and apply them to solve practical problems, such as volunteer training mini-program,arxiv paper query,weather query, etc.
- Proficient in multiple programming languages, frameworks, and productivity tools like Cursor and DeepWiki to efficiently manage project workflows. A strong communicator with a team-oriented mindset, actively sharing knowledge and experience through personal blogs and websites to organize technical expertise.
- With strong learning ability, I can track industry developments through reading academic papers and technical documents. I am passionate about participating in open source community projects, such as building WeChat scheduled message push delivery,news crawling, submitting and merging PRs for blink.cmp, writing plugins for Wox, and developing local LLM applications Agentic-RAG.